
Final Report: Computing Optimal Regularizers for Online Linear Optimization

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Abstract

This report surveys the paper *Computing Optimal Regularizers for Online Linear Optimization* by Gatmiry, Schneider, and Jegelka (COLT 2025), which addresses a long-standing question in online learning: given a specific online linear optimization (OLO) instance with convex symmetric action set $\mathcal{X}$ and loss set $\mathcal{L}$, can one algorithmically compute a regularizer that makes Follow-The-Regularized-Leader (FTRL) achieve the instance-optimal regret up to universal constants? The authors answer this affirmatively. They first strengthen the universality result of Srebro et al. (2011) by removing a spurious $\log T$ factor, proving that some FTRL instance always matches the minimax rate $\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}$ up to a universal constant. They then give a preprocessing algorithm that, using membership and linear-optimization oracles, outputs such a regularizer; the algorithm is fully polynomial for constant dimension d . A matching NP-hardness lower bound for certifying strong convexity complements these upper bounds. We summarize the problem setup, the main theorems, and the key proof ideas—most notably the quasi-quadratic approximation and the convex program used to search over regularizers.

1 Introduction

Online learning studies sequential decision making under adversarial loss: over T rounds, a learner chooses actions $x_t \in \mathcal{X}$ and suffers losses determined by an adversary. A central special case covered in our lectures is Online Convex Optimization (OCO), and its linear specialization, Online Linear Optimization (OLO), where at each round the learner picks $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$, the adversary reveals $\ell_t \in \mathcal{L} \subseteq \mathbb{R}^d$, and the learner suffers linear loss $\langle x_t, \ell_t \rangle$ [6, 7]. Performance is measured by the regret

$$\text{Reg}_T = \sum_{t=1}^T \langle x_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x, \ell_t \rangle.$$

OLO may look deceptively simple, but it is in fact a universal primitive: since any OCO loss can be linearized at the played point via its subgradient, OLO algorithms immediately yield OCO algorithms with the same regret bound [7]. OLO also subsumes adjacent problems such as Blackwell approachability and extensive-form equilibrium computation [1].

The standard approach to OLO is Follow-The-Regularized-Leader (FTRL), where the learner plays

$$x_t = \arg \min_{x \in \mathcal{X}} \left\{ \eta \sum_{s < t} \langle x, \ell_s \rangle + f(x) \right\}$$

for some convex regularizer $f : \mathcal{X} \rightarrow \mathbb{R}$ [7]. Although FTRL achieves $O(\sqrt{T})$ regret in great generality, the constants and dimension dependence depend dramatically on f . The classical experts example is instructive: on the simplex, FTRL with $f(x) = \frac{1}{2}\|x\|_2^2$ yields $O(\sqrt{dT})$ regret, while

negative entropy yields $O(\sqrt{T \log d})$ —an exponential improvement in d . For other geometries, quadratic regularization is itself optimal [1]. Historically this choice has been made by humans on a case-by-case basis: entropies for the simplex, log-determinants for spectraplexes [5], p -norms for general balls, and self-concordant barriers for bandits.

The paper we study, **Computing Optimal Regularizers for Online Linear Optimization** by Garmiry, Schneider, and Jegelka [1], asks the following question. *Given a specific OLO instance $(\mathcal{X}, \mathcal{L})$, can we algorithmically compute a regularizer f such that FTRL with f achieves regret within a universal constant of the minimax optimum for that instance?* The prior state of the art was the universality result of Srebro et al. (2011), which used martingale-type arguments to show non-constructively that some regularizer makes FTRL near-optimal up to an extra $\log T$ factor. The paper makes three main advances. First, a tight existence theorem (Theorem 2 of [1]) shows that for any convex symmetric $(\mathcal{X}, \mathcal{L})$, some FTRL instance achieves regret $O(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T})$, removing the spurious $\log T$ factor and establishing instance-optimality up to a universal constant. Second, a constructive algorithm, given membership and linear-optimization oracles for $\mathcal{X}$ and $\mathcal{L}$, outputs such a regularizer with preprocessing cost $\exp(O(d^2 \log d))$ and can then be run efficiently online; for constant d , this is fully polynomial. Third, a matching NP-hardness result shows that deciding whether a given convex function is α -strongly convex with respect to a given norm is NP-hard, so some form of computational overhead is unavoidable.

These results tie directly to the material from class: FTRL and its stability analysis [7], strong convexity and Bregman divergences, and the standard regret decomposition separating a stability term from a regularization penalty. The paper’s contribution is to turn the existential, geometry-dependent guarantees of prior work into a genuinely algorithmic statement: universality of FTRL not only holds, it can be computed. Section 2 formalizes the problem setup. Sections 3 and 4 describe the main theorems and distill selected proofs. Section 5 concludes.

2 Problem Setup

We now formalize the OLO problem, following the notation of [1] and adapting it to the notation used in lectures.

The OLO game. Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the action set and $\mathcal{L} \subseteq \mathbb{R}^d$ the loss set. At each round $t = 1, \dots, T$, the learner selects an action $x_t \in \mathcal{X}$ (possibly randomized and depending on the history), the adversary selects a loss vector $\ell_t \in \mathcal{L}$, and the learner incurs linear loss $\langle x_t, \ell_t \rangle$ and observes ℓ_t (full-information feedback). The regret against the best fixed action in hindsight is

$$\text{Reg}(\mathbf{x}, \ell) = \sum_{t=1}^T \langle x_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x, \ell_t \rangle. \quad (1)$$

Structural assumptions. The paper imposes two standing assumptions on $\mathcal{X}$ and $\mathcal{L}$ [1]. Both sets are closed, bounded convex bodies in $\mathbb{R}^d$ that contain a ball around the origin and are contained in a ball, so that the associated gauges are actual norms. Both are also centrally symmetric, i.e., $\mathcal{X} = -\mathcal{X}$ and $\mathcal{L} = -\mathcal{L}$. This covers essentially all geometric OLO instances of interest (balls, ellipsoids, hypercubes, spectraplexes, and so on). Under these assumptions, each symmetric convex body $\mathcal{C}$ induces a Minkowski norm

$$\|v\|_{\mathcal{C}} = \inf\{t > 0 : v \in t \cdot \mathcal{C}\}. \quad (2)$$

In particular, $\|\cdot\|_{\mathcal{X}}$ is the norm with unit ball $\mathcal{X}$, and $\|\cdot\|_{\mathcal{L}^\circ}$, where $\mathcal{L}^\circ$ is the polar of $\mathcal{L}$, is the norm dual to the loss geometry. These form the primal/dual pair used throughout the regret analysis.

Minimax regret and optimal rate. The minimax regret of the instance $(\mathcal{X}, \mathcal{L})$ is $R_T^*(\mathcal{X}, \mathcal{L}) = \inf_{\text{Alg}} \sup_{\ell \in \mathcal{L}^T} \mathbb{E}[\text{Reg}(\mathbf{x}, \ell)]$, where the infimum is over all (possibly randomized) algorithms. Under the assumptions above this quantity grows like $\sqrt{T}$, and we define the instance-dependent optimal rate as

$$\text{Rate}(\mathcal{X}, \mathcal{L}) = \limsup_{T \rightarrow \infty} \frac{R_T^*(\mathcal{X}, \mathcal{L})}{\sqrt{T}}. \quad (3)$$

This is the fundamental geometric quantity the paper aims to match: it depends only on $\mathcal{X}$ and $\mathcal{L}$ and captures exactly the dimension-dependent leading constant. For the simplex with ℓ_∞ losses

one has $\text{Rate} = \Theta(\sqrt{\log d})$ (the entropy rate), while for the Euclidean ball with Euclidean losses $\text{Rate} = \Theta(1)$.

FTRL and its regret decomposition. The algorithmic framework under study is Follow-The-Regularized-Leader. Given a convex regularizer $f : \mathcal{X} \rightarrow \mathbb{R}$ and a learning rate $\eta > 0$, FTRL plays

$$x_t = \arg \min_{x \in \mathcal{X}} \left\{ \eta \sum_{s < t} \langle x, \ell_s \rangle + f(x) \right\}. \quad (4)$$

From our lecture notes [7], if f is 1-strongly convex with respect to some norm $\|\cdot\|$ (with dual $\|\cdot\|_*$), FTRL satisfies

$$\text{Reg}_T \leq \frac{\text{Range}(f)}{\eta} + \eta \sum_{t=1}^T \|\ell_t\|_*^2 \leq 2\sqrt{\text{Range}(f) T \max_{\ell \in \mathcal{L}} \|\ell\|_*^2}, \quad (5)$$

where $\text{Range}(f) = \max_{x \in \mathcal{X}} f(x) - \min_{x \in \mathcal{X}} f(x)$, and the second inequality follows by optimizing η . This bound makes the design problem transparent: to match $\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}$, one must minimize the product $\text{Range}(f) \cdot \max_{\ell \in \mathcal{L}} \|\ell\|_*^2$ subject to f being 1-strongly convex with respect to the norm whose dual measures the losses. Choosing that norm, and then choosing f strongly convex with respect to it, is precisely what the paper makes algorithmic.

Scope of this report. The paper proves many results, including extensions to various oracle models. For this report we focus on three results most closely tied to the course: the existence of an optimal-rate FTRL regularizer (a tightened Srebro et al. universality theorem), the constructive algorithm based on the quasi-quadratic local approximation, and the key proof ideas, including the convex program and the strong-convexity separation oracle.

3 Main Results

The paper shows that the search for a good FTRL regularizer can be made both information-theoretically tight and algorithmically constructive [1]. For any symmetric convex OLO instance $(\mathcal{X}, \mathcal{L})$, it identifies a regularizer whose regret matches the minimax benchmark $\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}$ up to universal constants, and then shows how to compute such a regularizer from oracle access to the geometry of $\mathcal{X}$ and $\mathcal{L}$.

Theorem 2 (existence of an ideal regularizer). There exists a regularizer f_0 such that FTRL with f_0 satisfies

$$\text{Reg}_T = O\left(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}\right).$$

This is the main information-theoretic result of the paper. Since $\text{Rate}(\mathcal{X}, \mathcal{L})$ is exactly the minimax $\sqrt{T}$ -scaled benchmark defined in Section 2, the theorem shows that FTRL is instance-optimal up to a universal constant. Compared with Srebro et al. [2], the new result removes the extra $\log T$ factor.

Theorem 3 (smooth near-optimal regularizer). The paper next proves that one may assume, without losing the regret guarantee, that the regularizer is smooth: there exists f such that FTRL with f still achieves

$$\text{Reg}_T = O\left(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}\right),$$

while f also has bounded first, second, and third derivatives [1]. This step is crucial algorithmically. The ideal regularizer f_0 is obtained by an existence argument and need not be well behaved enough for direct numerical optimization; Gaussian smoothing fixes this while preserving strong convexity.

Quasi-quadratic approximation. After smoothing, the paper approximates f over a finite ϵ -net $S = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ by patches of the form

$$q_i(x) = r_i + \langle v_i, x - x_i \rangle + \frac{1}{2}(x - x_i)^\top \Sigma_i (x - x_i) - \frac{M}{6} \|x - x_i\|_2^3, \quad g(x) = \max_{i \in [N]} q_i(x),$$

where (r_i, v_i, Σ_i) represent approximate values, gradients, and Hessians at the centers, and M is a Hessian-Lipschitz bound for f [1]. The cubic correction is essential: it turns a local Taylor approximation into a valid global lower model.

Theorem 4 (finite-dimensional convex program). For a fine enough discretization, the patch parameters can be chosen as a feasible point of an explicit convex program [1]. The resulting envelope g is convex, strongly convex with respect to $\|\cdot\|_{\mathcal{L}^\circ}$, and bounded on $\mathcal{X}$ by $O(\text{Rate}(\mathcal{X}, \mathcal{L})^2)$. Thus the infinite-dimensional search over convex regularizers is reduced to a finite-dimensional convex optimization problem over finitely many values, gradients, and Hessians.

Algorithmic pipeline. The constructive result can be summarized in five steps:

1. prove the existence of an ideal regularizer f_0 ;
2. smooth f_0 to obtain a well-behaved regularizer f ;
3. discretize $\mathcal{X}$ and approximate f by local quasi-quadratic patches;
4. optimize the patch parameters via a convex program, solved with cutting planes and a separation oracle derived from linear optimization over $\mathcal{L}$;
5. run ordinary FTRL online with the computed regularizer g .

The online algorithm itself is standard FTRL; the novelty is that the regularizer is now computed from the problem geometry [1].

Main algorithmic theorem. Combining the previous ingredients, the paper shows that one can compute a regularizer g that is strongly convex with respect to $\|\cdot\|_{\mathcal{L}^\circ}$ and bounded by $O(\text{Rate}(\mathcal{X}, \mathcal{L})^2)$ on $\mathcal{X}$. Plugging this g into standard FTRL yields

$$\text{Reg}_T = O\left(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}\right).$$

The preprocessing cost is exponential in the dimension d , whereas the online stage is efficient; for constant d , the entire procedure is fully polynomial [1].

Motivating example and caveat. A useful motivating example is the experts problem: quadratic regularization leads to rate $\Theta(\sqrt{d})$, while negative entropy achieves the optimal $\Theta(\sqrt{\log d})$ [1]. This illustrates why geometry-matched regularizers matter. At the same time, the constructive theorem of the paper is stated for symmetric convex bodies, so the simplex should be read as motivation rather than as a literal running instance of the oracle-based construction.

Theorem 5 (hardness). The paper also proves a computational lower bound showing that certifying strong convexity relative to the unknown loss geometry is hard in general [1]. This is not the theorem behind regret optimality; instead, it explains why the offline preprocessing can require exponential dependence on d .

4 Proofs

We focus on three proof ideas that drive the paper: the existence proof for the ideal regularizer, the Gaussian smoothing step, and the finite-dimensional approximation argument. Our goal is not to reproduce every technical detail, but to isolate the main lemmas and the role each one plays in the overall construction.

4.1 Proof sketch of Theorem 2

Equation (5) from the lecture notes [7] already identifies the core design problem: to obtain regret of order $\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}$, it is enough to construct a regularizer that is strongly convex with respect to $\|\cdot\|_{\mathcal{L}^\circ}$ and has range $O(\text{Rate}(\mathcal{X}, \mathcal{L})^2)$ on $\mathcal{X}$.

The paper builds such a regularizer by sharpening the universality argument of Srebro et al. [2]. Their analysis relates $O(C\sqrt{T})$ learnability to a martingale-type inequality adapted to the geometry of the instance, and then converts this inequality into a regularizer. The reason their bound loses a $\log T$ factor is that the conversion is carried out for an exponent $p < 2$, introducing a factor of order $1/(2-p)$; choosing p close to 2 to recover a $\sqrt{T}$ rate then creates the extra logarithm. The key technical improvement in [1] is to prove the needed estimate directly at the endpoint $p = 2$.

Once this endpoint martingale bound is available, Lemma 7 in the paper produces a 2-homogeneous convex function f_0 such that f_0 is 1-strongly convex with respect to $\|\cdot\|_{\mathcal{L}^\circ}$ and

$$\max_{x \in \mathcal{X}} |f_0(x)| = O(\text{Rate}(\mathcal{X}, \mathcal{L})^2).$$

Substituting f_0 into (5) immediately gives $\text{Reg}_T = O(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T})$. Since $\text{Rate}(\mathcal{X}, \mathcal{L})$ is the minimax benchmark from Section 2, this is exactly the desired notion of optimality up to universal constants.

4.2 Proof sketch of Theorem 3

The regularizer f_0 from Theorem 2 is optimal from a regret point of view, but it need not be smooth enough for the later finite-dimensional approximation. The paper therefore defines a Gaussian-smoothed regularizer

$$f(x) = \mathbb{E}[f_0(x + \sigma Z)], \quad Z \sim \mathcal{N}(0, I)$$

[1].

The first observation is structural: convexity and strong convexity are preserved under expectation. Every shifted copy $x \mapsto f_0(x + \sigma z)$ has the same curvature as f_0 , so averaging over z keeps the strong-convexity parameter with respect to $\|\cdot\|_{\mathcal{L}^\circ}$ unchanged. Thus smoothing does not damage the regret guarantee.

The second observation is analytic. By differentiating under the expectation, all derivatives of f can be written in terms of Gaussian moments multiplied by values of f_0 . Because f_0 is 2-homogeneous and has controlled growth on $\mathcal{X}$, these Gaussian moment formulas yield explicit bounds on the first, second, and third derivatives of f [1]. Choosing the smoothing scale σ appropriately balances these bounds, so one obtains a regularizer that is still near-optimal but now has controlled local geometry. This is exactly what makes the patch-based approximation possible.

4.3 Proof sketch of Theorem 4

The proof of Theorem 4 has three conceptual steps.

First, since the smoothed regularizer f has Lipschitz Hessian, its second-order Taylor expansion with a cubic correction gives a global lower model around every center x_i :

$$q_i(x) = f(x_i) + \langle \nabla f(x_i), x - x_i \rangle + \frac{1}{2}(x - x_i)^\top \nabla^2 f(x_i)(x - x_i) - \frac{M}{6}\|x - x_i\|_2^3.$$

The cubic term is the crucial innovation here: without it, the quadratic Taylor approximation would only be locally accurate, while with it the patch stays below f globally [1].

Second, after choosing an ϵ -net $S = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$, the paper introduces variables (r_i, v_i, Σ_i) meant to represent the value, gradient, and Hessian of the unknown regularizer at each net point. The true triples $(f(x_i), \nabla f(x_i), \nabla^2 f(x_i))$ satisfy the constraints of the convex program up to the discretization slack, so the smooth near-optimal regularizer itself certifies feasibility [1].

Third, one must show that the envelope $g(x) = \max_i q_i(x)$ keeps enough curvature. A maximum of strongly convex functions is not automatically strongly convex, because the active branch could switch in a harmful way. The convex program prevents this through interpolation constraints that force any active patch at x to come from a nearby center. Nearby patches inherit the curvature of f , and the matrices Σ_i are constrained to be curved in every loss direction. Consequently, the active patch is locally strongly convex at every point, which implies that g is strongly convex with respect to $\|\cdot\|_{\mathcal{L}^\circ}$ [1]. At the same time, the objective of the convex program controls $\sup_{x \in \mathcal{X}} g(x)$, so the range of g remains of the same order as that of the smooth optimal regularizer.

Finally, the remaining computational step is to solve the convex program. The paper shows that its constraints admit a separation oracle based on linear optimization over $\mathcal{L}$, which allows cutting-plane methods to recover a computable regularizer g [1]. Plugging this g into FTRL completes the construction.

Remark on Theorem 5. Theorem 5 is a hardness result about the preprocessing stage, not the theorem that proves optimal regret. The near-optimal regret bound comes from Theorems 2–4 and

the final algorithmic theorem; Theorem 5 only explains why, in high dimension, one should not expect the certification step to be easy [1].

5 Conclusion

The paper of Garmory, Schneider, and Jegelka advances the theory of FTRL from the qualitative universality of Srebro et al. (2011) to a quantitative, algorithmic universality. Three messages stand out. First, FTRL is instance-optimal without logarithmic slack: for any convex symmetric $(\mathcal{X}, \mathcal{L})$, some FTRL regularizer matches the minimax optimal $\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T}$ up to a universal constant, resolving a long-standing near-miss in the OLO literature. Second, optimal regularizers can be computed: given oracle access to $\mathcal{X}$ and $\mathcal{L}$, the quasi-quadratic local-approximation scheme reduces regularizer design to a tractable convex program, polynomial in constant dimension, giving the first general-purpose “compiler” from a description of the OLO geometry to a near-optimal FTRL algorithm. Third, some computational cost is unavoidable: the NP-hardness of certifying α -strong convexity shows that, outside constant dimension, one cannot hope to certify regularizer quality without exponential blow-up, justifying the paper’s preprocessing cost.

Beyond concrete regret bounds, the conceptual contribution is to recast regularizer design as a convex optimization problem over the space of convex functions, approximated via a discretized, second-order parametric family. This bridge between FTRL analysis and tractable numerical optimization suggests further applications to adaptive, instance-dependent, and bandit variants of online learning.

6 Open Questions

While the constructive algorithm by Garmory, Schneider, and Jegelka elegantly resolves the computability of optimal regularizers, its reliance on a dense ϵ -net introduces a preprocessing cost exponential in the dimension d . This computational bottleneck naturally opens several promising avenues for future research, shifting the focus from general computability to practical tractability.

Exploiting Structured Geometries. The current framework makes minimal assumptions, requiring only a separation oracle for generic symmetric convex bodies $(\mathcal{X}, \mathcal{L})$. However, in practice, loss sets and action spaces often possess highly structured geometries (e.g., ℓ_p -balls, hypercubes, or intersections of simple half-spaces). A major open question is whether one can substantially reduce the reliance on the current patch-based construction for these specific cases. Can we leverage the explicit algebraic boundaries of structured geometries to derive useful local curvature information, or simplified Hessian surrogates, without resorting to a brute-force spatial discretization?

Adaptive Discretization and Active Sets. The algorithm’s reliance on a uniform spatial grid is the primary source of the curse of dimensionality. A natural alternative would be to use adaptive or multiscale discretization. Instead of a uniform ϵ -net, can we design an active-set method that only samples patch centers x_i in regions of high curvature or near the boundary of the feasible domain? Developing a sparse active-patch representation—where the regularizer is parameterized by a minimal subset of critical support points—could drastically reduce the size of the convex program while preserving the global strong convexity guarantee.

The Sub-Optimal Approximation Trade-off. Theorem 5 establishes the NP-hardness of certifying exact α -strong convexity relative to an arbitrary dual norm. This implies a strict barrier for exact optimal bounds. A related question is whether one can efficiently compute a mildly sub-optimal regularizer, for example, one achieving regret within a controllable dimension-dependent factor of $O(\text{Rate}(\mathcal{X}, \mathcal{L})\sqrt{T})$. This would ask whether one can trade exact optimality for substantially better preprocessing time in high dimensions, a crucial step for bridging the gap between this theoretical milestone and large-scale deployment in online learning tasks.

These directions are closely related: all three ask whether the current patch-based construction can be compressed without losing the two properties that make the final FTRL guarantee go through, namely global strong convexity and a controlled range bound. In the current proof, these are obtained through a uniform discretization of $\mathcal{X}$, worst-case locality of the active patch, and explicit

convex-program constraints on the local Hessian surrogates. Any substantial improvement in run-time therefore seems to require a more economical geometric representation of the regularizer, rather than a simple implementation-level optimization of the existing framework.

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